Ring osc on VGA (tt_um_algofoogle_vgaringosc): silicon measurements

Tiny Tapeout (TT) GF180MCU shuttle ttgf0p2, mux address 614 · Measured by Matt Venn, 2026-11-09
Project page: https://tinytapeout.com/chips/ttgf0p2/tt_um_algofoogle_vgaringosc

Chip 1 (ROM commit ff701baa) unless noted. Setup: TT demo board TTDBv3 (RP2350 microcontroller, TT SDK 3.1.1), ttgf0p2 carrier v0.1, VDD_CORE = 3.300 V (demo-board 3V3 through 0 Ω links; carrier regulator U1 not fitted), project clock 25.25 MHz from the RP2350, room temperature (not controlled). Keysight HD304MSO oscilloscope, 10:1 probe, 3.2 GSa/s. Ring frequency = 8 × median of 30 single-period readings of the ÷8 output uio[7]; each configuration run 3 times with a project reset between. Check: clksel = 0 gives 25.2524 MHz (0.01 % from the RP2350 clock), and ÷8 × 8 agrees with direct raw-pin readings within 0.25 %.

Result: the rings run from 23.7 MHz (161 stages) to 220.5 MHz (3 stages); the fixed 5× inv_1 ring runs at about 320 MHz. Period fits T = 0.240 ns × N + 4.19 ns: about 120 ps per inv_2 stage (T = 2·N·t_pd) plus about 4.2 ns fixed loop overhead (tap multiplexer and clock buffer). Your docs expected about 15 to 350 MHz.

clksel	stages N	f (MHz)	restart spread %	fit resid. (ns)
2	3	220.51	0.03	-0.38
3	5	196.22	0.02	-0.30
4	7	176.81	0.09	-0.22
5	9	160.66	0.04	-0.13
6	13	143.51	0.05	-0.34
7	17	125.52	0.14	-0.31
8	21	108.94	0.03	-0.05
9	25	98.28	0.04	-0.02
10	33	77.69	0.16	+0.76
11	41	69.36	0.03	+0.39
12	49	60.25	0.16	+0.65
13	65	49.87	0.03	+0.26
14	97	35.96	0.17	+0.34
15	161	23.72	0.04	-0.66
10*	5 (inv_1)	319.6	1.3	–

Table 1. Median ring frequency per tap (3 restarts). *clksel2 = 1, clksel = 10. Spread = (max − min) / mean of the 3 restart medians.

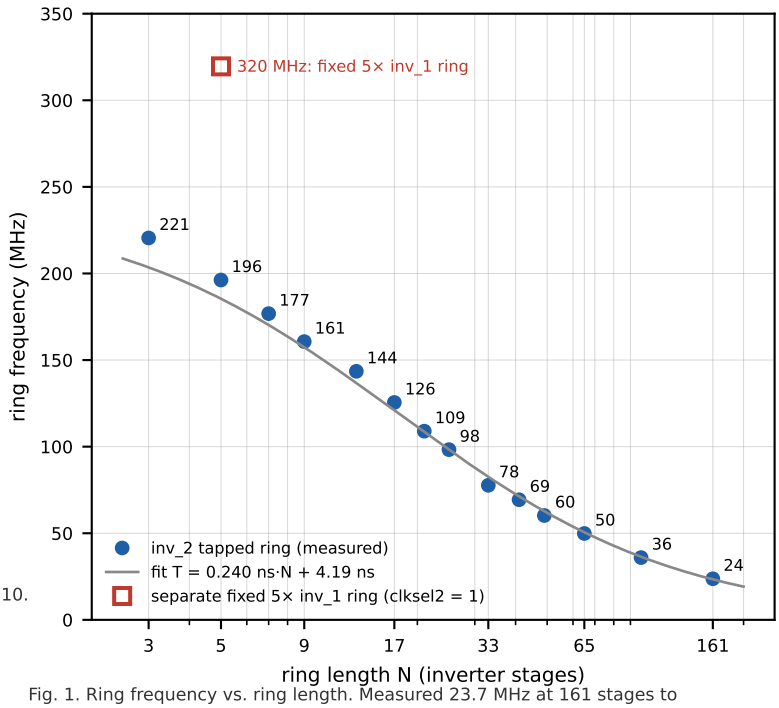


Fig. 1. Ring frequency vs. ring length. Measured 23.7 MHz at 161 stages to 220.5 MHz at 3 stages; fit residuals -0.66 to +0.76 ns (ring segments differ). Red square: separate fixed 5× inv_1 ring (clksel2 = 1), 320 MHz, not part of the tapped ring.



Fig. 2. Raw pin uio[4], clksel = 15 (161 stages), 20 ns/div. Scope mean 23.728 MHz, s.d. 13.8 kHz over 12.3 k cycles.

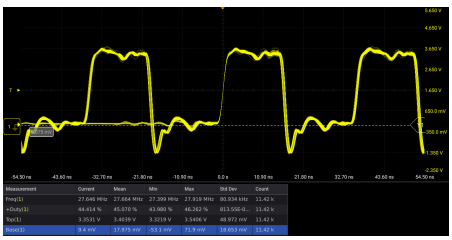


Fig. 3. ÷8 output uio[7], clksel = 2 (3 stages), 10.9 ns/div. Scope mean 27.66 MHz ×8 = 221.3 MHz over 11.4 k cycles.

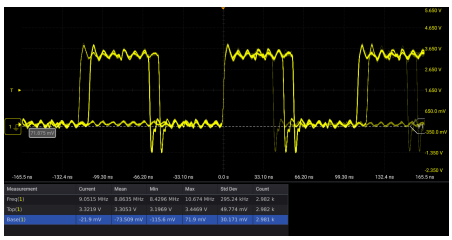


Fig. 4. ÷8 output, 161 stages entered from clksel = 2 without reset: 9.05 MHz ×8 = 72.4 MHz, the 3rd-harmonic mode (3.06× the 23.7 MHz fundamental).

Notes / limits

- Multi-mode locking: switching to the 161-stage ring without a reset gave the 3rd-harmonic mode (about 72.5 MHz, 3.06× the fundamental) in 3 of 8 tries, plus once earlier the same day. With a reset after each change: 12/12 fundamental (23.67 to 23.70 MHz). Advice for users: reset after changing clksel.
- 5× inv_1 ring (320 MHz) is the least certain point: single readings spread widely at this speed, and the 3-bit counter clocked at a 3.1 ns period may be marginal.
- Raw pin uio[4] works only up to about 50 MHz with a standard probe (it agrees with ÷8 × 8 within 0.25 % for clksel 13 to 15); the 5× inv_1 ring does not toggle it at all. Pad edges at 3.3 V: rise 1.82 ns, fall 1.19 ns (10 to 90 %).
- The ÷2/÷4/÷8 counter is held in reset during VGA horizontal blanking, so the ÷8 output has a gap each line; single-period readings near the trigger are not biased by it.
- Pin fault on chip 1: the uio[1] pad (clksel2[1]) is shorted to ground. The RP2350 cannot pull it high, the same RP2350 pin toggles with the carrier removed, and the bond wires look intact. So clksel2 = 2 and 3 were measured on chip 2 only. Tip for TT SDK users: after enabling the project, uio[0]/uio[1] stayed RP2350 inputs despite uio_oe_pico = 0b11; set them with Pin.OUT.
- Chip 2 (second carrier, ROM commit 7416232b): only ui_in[1], ui_in[7] and uio[0..3] reach the chip (the other inputs are held low), so only the 3-stage tap, the fixed rings and the clock paths could be selected. 3 stages: 216.0 MHz (chip 1: 220.5 MHz, −2.0 %). 5× inv_1: 311.6 MHz (chip 1: 319.6 MHz, −2.5 %). clksel2 = 3 (!clk): 25.252 MHz, as expected. 5× inv_4: median 206.5 MHz over 3 restarts, but the ÷8 output is irregular (duty 23 to 92 %), so the counter may be miscounting; treat this value as unconfirmed.
- Not tested: VGA output (no VGA PMOD or monitor), altclk path with a clock (with ui_in[4] static there are no edges, as expected). Full tap sweep on chip 1 only; one supply voltage, room temperature only.

Design source: github.com/algofoogle/ttgf0p2-vga-ring-osc (commit cdefb8e).
Raw data and scope captures: https://github.com/mattvenn/ttgf0p2-vgaringosc-measurements

Made with Claude